How Weather Affects Transportation

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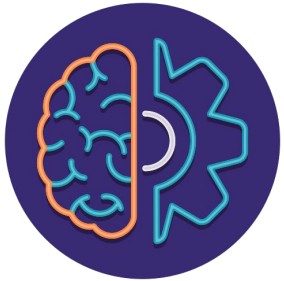
Problem Statement

We want to see…

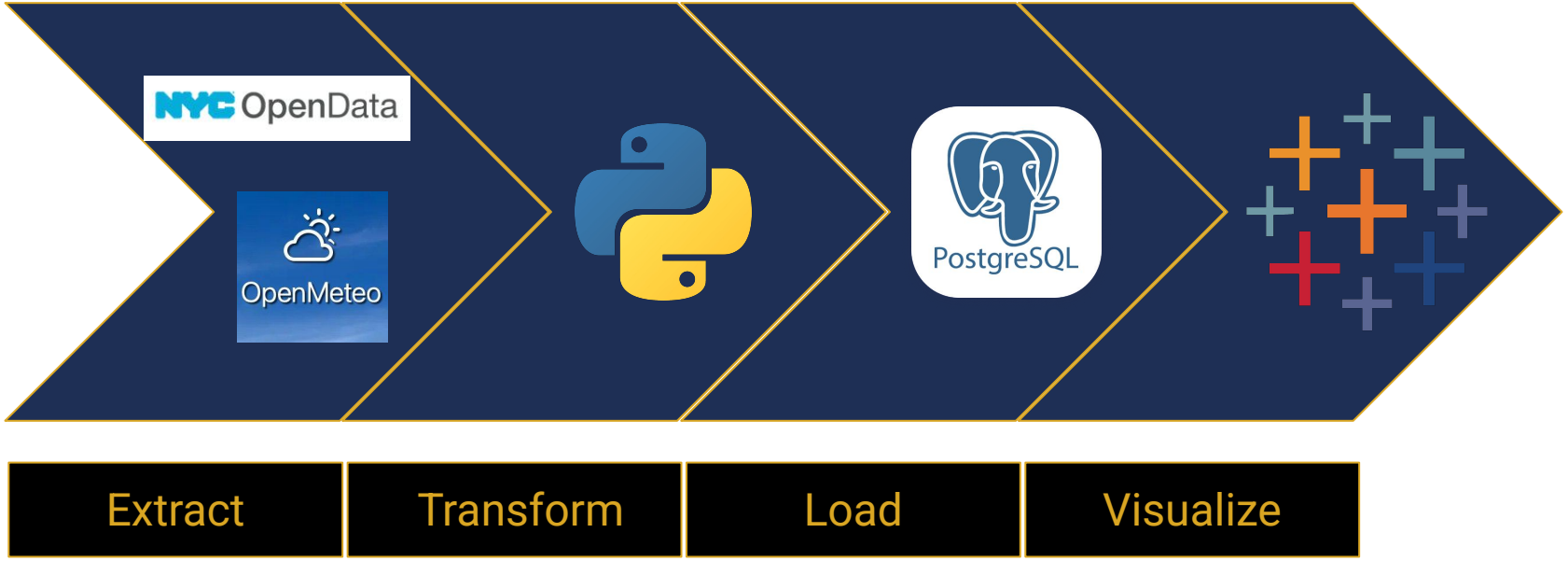
1. How MTA ridership and traffic volume changes depending on different weather conditions

2. What weather conditions have the greatest impact

3. If we can accurately predict these changes in the future with machine learning



Architecture



Datasets & APIs

Weather Source Open-Meteo API

Data Type: Temperature, rain, snowfall, wind speed

Coverage: NYC (hourly) 2020–2024

Use Case: Primary weather input for all models

Bridges & Tunnels

Data Type: E-ZPass and toll vehicle counts

Coverage: MTA (hourly) 2010–2025

Use Case: Analyze impact of weather on bridge/tunnel traffic

Traffic Volume

Data Type: Vehicle counts by street and direction

Coverage: All 5 boroughs (hourly) 2000–2026

Use Case: Key target variable for congestion models

Bus Ridership

Data Type: Ridership and transfers by route

Coverage: MTA (hourly) 2020–2024

Use Case: Identify shifts from driving to bus transit during poor weather

Subway Ridership

Data Type: Total station entries

Coverage: MTA (hourly) 2020–2024

Use Case: Correlate subway demand spikes with rain and snow events

Seamless Collaboration with Docker

- **The Problem:** Coordinating environments across a 6-person team risks version conflicts and “it works on my machine” errors.
- **Solution:** Containerizing our PostgreSQL database and Python environment using Docker.
- **Why?:** Ensures the entire stack runs instantly on any machine with a single `docker-compose up` command, allowing us to focus on the data, not setup.

docker-compose.yml

```yaml
version: '3.8'

services:
  db:
    image: postgres:latest
    restart: always
    environment:
      POSTGRES_DB: mta_weather_db
      POSTGRES_USER: admin
      POSTGRES_PASSWORD: password
    ports:
      - "5432:5432"
    volumes:
      - pgdata:/var/lib/postgresql/data

  app:
    build: .
    volumes:
      - .:/app
    # This automatically loads our API key from the .env file
    env_file:
      - .env
    depends_on:
      - db

volumes:
  pgdata:
```

Why SQL?

- Our data is structured which SQL is excels at
- We have multiple datasets which we need to join together and SQL provides the tools and queries necessary to do this
- The schema of our data is defined and strict
- Cloud storage database management systems often use SQL such as Supabase which is something that fits our needs for this project.

PostgreSQL

- PostgreSQL as the 'brain' of our stack. While Supabase makes it easy for us to access our data.
- The Postgres engine underneath allows us to run the complex aggregations needed for certain models such as linear.
- Has the reliability needed to handle millions of rows provided by the MTA and NYC open data

The pivot to Supabase

- Supabase is an alternative to docker
- It’s a free web hosting platform which includes a fully managed PostgreSQL
- Easier to access and familiarize with
- Allows us move our project to our own server or to a different cloud provider in the future

Tableau for Visuals

- Tableau connects live to our PostgreSQL database (Supabase)

- Python writes master_hourly table for cleaned & merged data
- Stored in Supabase PostgreSQL

- Tableau connects via Native PostgreSQL connector
- Connect → To a Server → PostgreSQL

Live Connection

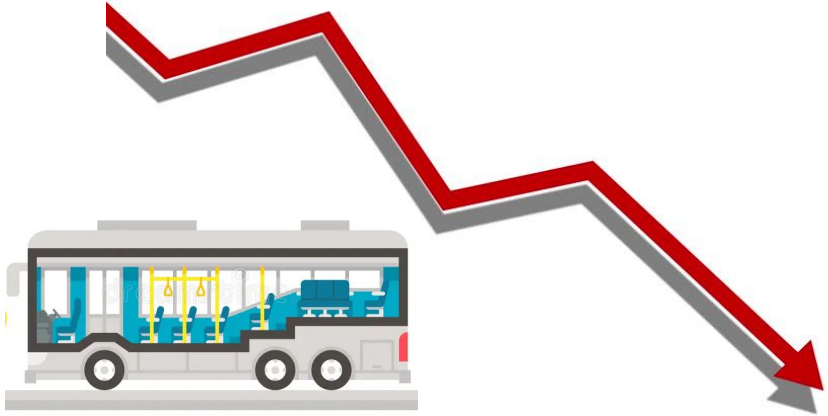
Pipeline run and Tableau visualization are connected end to end so the dashboard reflects whatever Python just wrote to the database

Models and Expected Insights

Linear Regression - Simple to interpret, by how much does MTA ridership and traffic volume change after say 1 inch of rain or snow

Logistic Regression - Given the forecast, is there a >70% chance of the MTA reporting high ridership.

Poisson Regression - 1 inch of rain reduces ridership by 15%



Thank You!

Any Questions?